

# Robustness of Hebrew Sentiment Analysis

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# Motivation: Variation and Fragility

- The same word/sentence has many surface forms a reader treats as identical.
- A model that flips its answer on a re-spelling is **\*\*fragile\*\*** - and clean accuracy hides it.
- תכנית ↔ תוכנית (full / defective spelling)
- בבית → ב בית (prefix segmentation)
- הכול → הכל (with / without niqqud)



# Research Questions

Q1

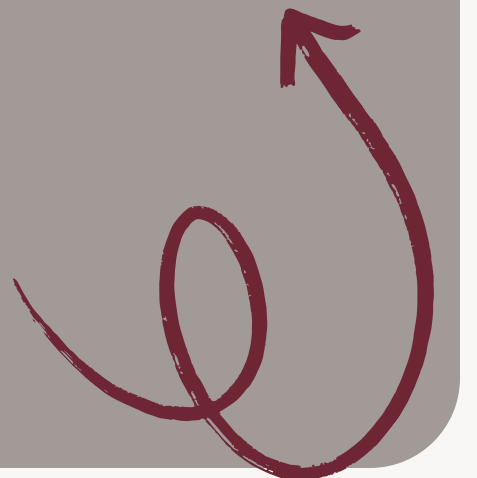
What method is most **robust** to meaning-preserving Hebrew variation?

Q2

Do **Hebrew vs. English** instructions change robustness?

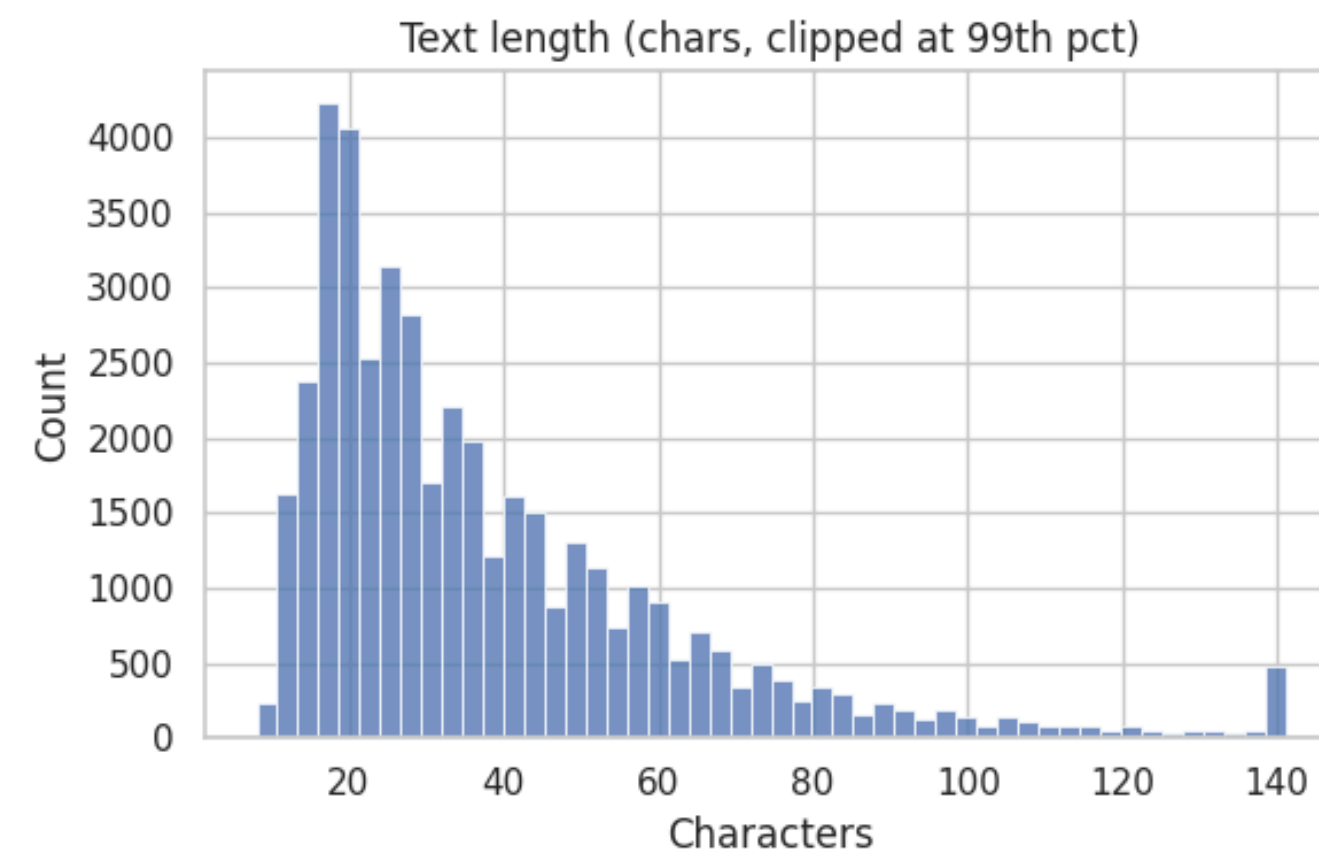
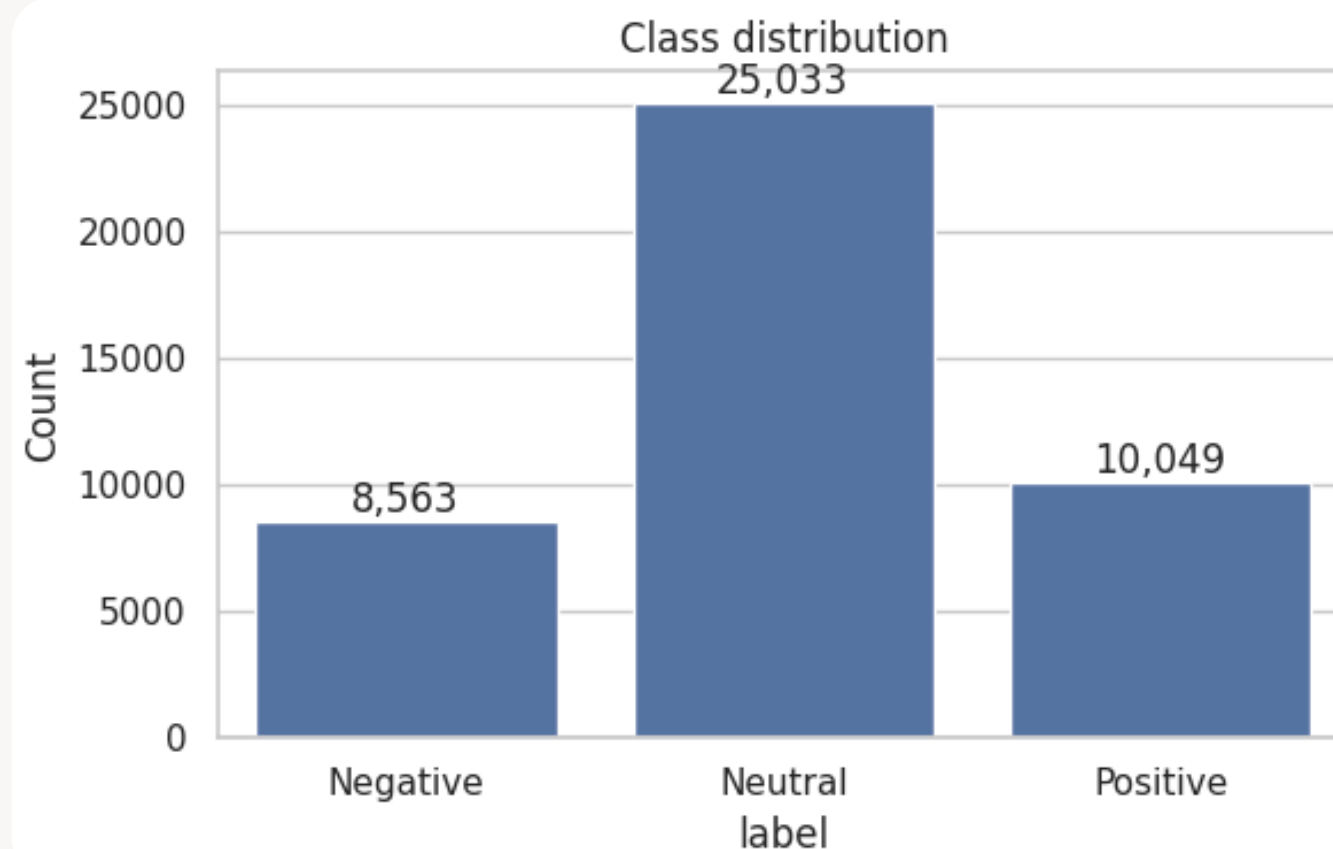
Q3

Does **tokenization** explain the fragility?



# Data And Evaluation Setup

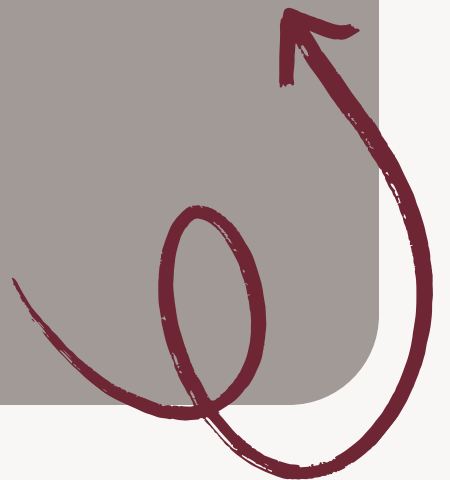
- `HebArabNlpProject/HebrewSentiment` - 43,645 texts, 3 classes.
- **Skewed:** Neutral 25k · Positive 10k · Negative 8.6k.
- Frozen **class-balanced 500-example** test subset - every method scored on the same rows.
- Texts are short (median 30 chars) → single-word edits matter.



# Evaluation Methods

- 01** Zero-shot - Claude Haiku 4.5, HE + EN prompts
- 02** Few-shot - 9 curated examples (3/class), HE + EN
- 03** RAG - retrieved few-shot via neodictabert embeddings, HE + EN
- 04** LoRA - fine-tuned DictaBERT (r=8, only 0.16% of params trained)

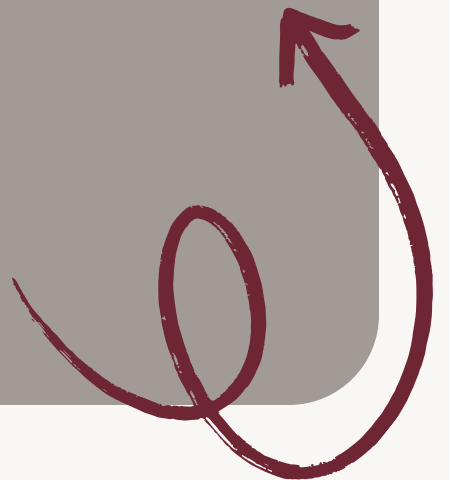
3 prompting methods × 2 languages + LoRA = **7 conditions**, same 500 rows.



# Meaning-Preserving Perturbations

- 01 Spelling** - full ↔ defective - תכנית ↔ תוכנית
- 02 Niqqud** - add vowel points (Dicta Nakdan) - שלום → שְׁלוֹם
- 03 Prefix** - detach clitic - בבית → ב בית
- 04 Typos** - final-form / typos - חלום → חלומ

Each edit carries a `changed` flag - metrics use only rows that actually changed



# Measuring Robustness

## Key Metrics for Evaluation

### Macro F1

Average of the per-class F1 scores - so the majority Neutral class can't hide poor Negative/Positive performance.

### Robustness Gap

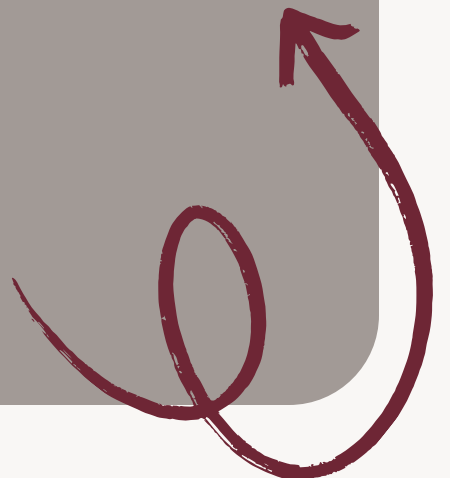
drop in Macro-F1 from clean → perturbed.  
Smaller = more stable

### Flip-Rate

of clean-correct predictions, the fraction that change.  
**No gold labels needed.**

### Tokenizer fertility

how many extra tokens an edit creates  
**(DictaBERT vs. tiktoken vs. Claude).**



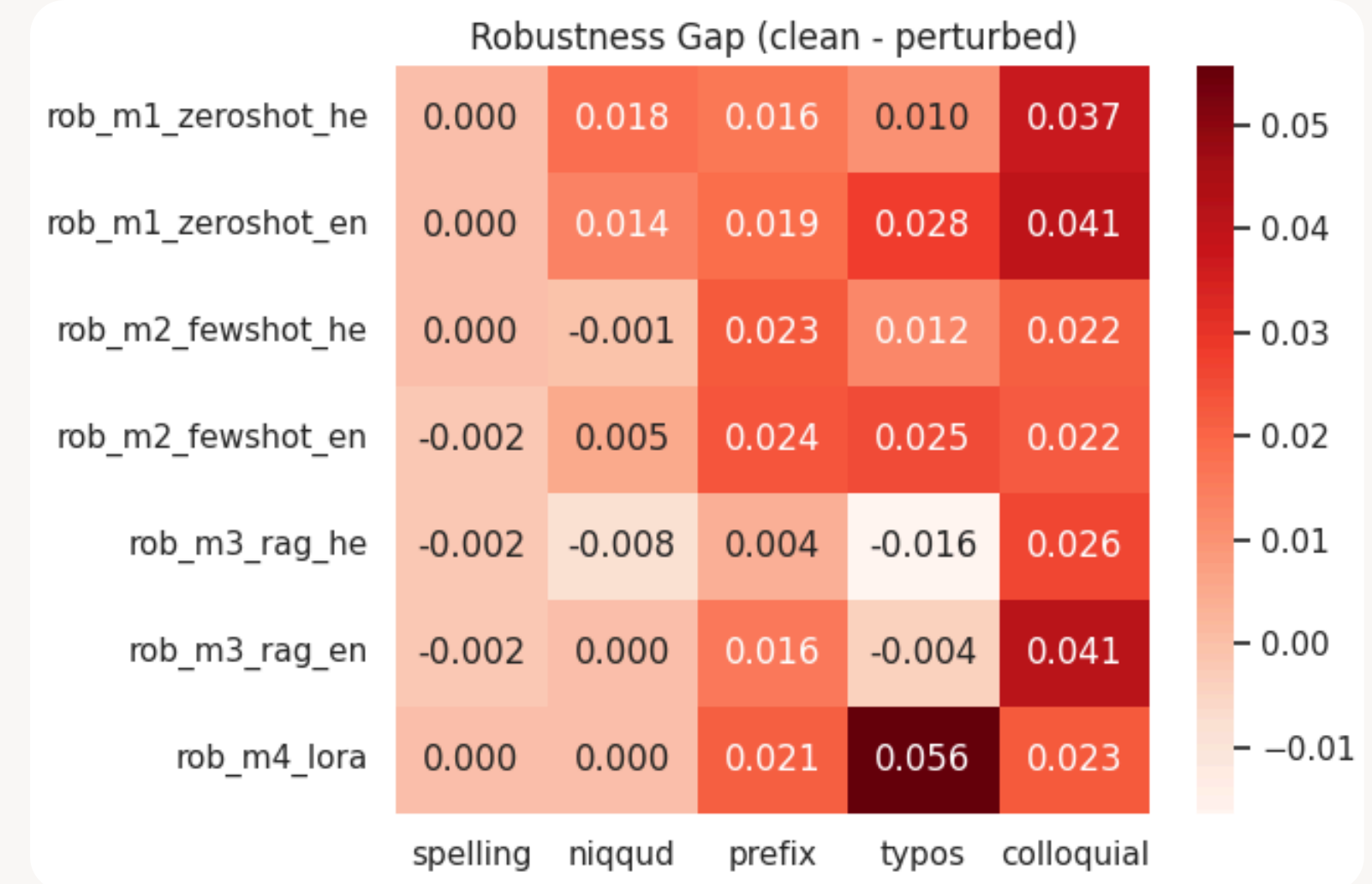
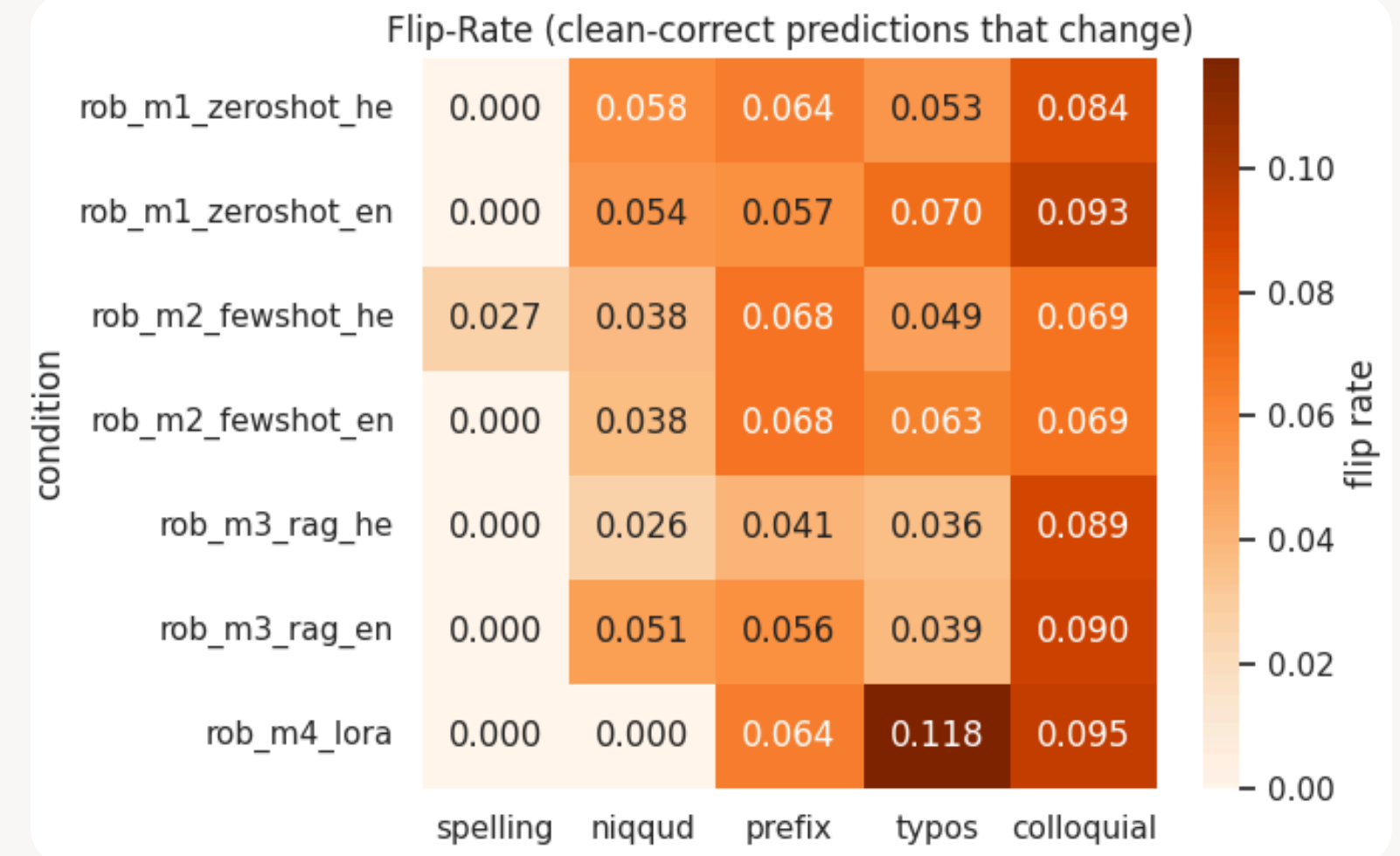
# Results: Robustness Gap and Flip-Rate

## Takeaways:

1. RAG rows are the “coolest” (most robust)
2. LoRA × typos cell is the hotspot in both panels

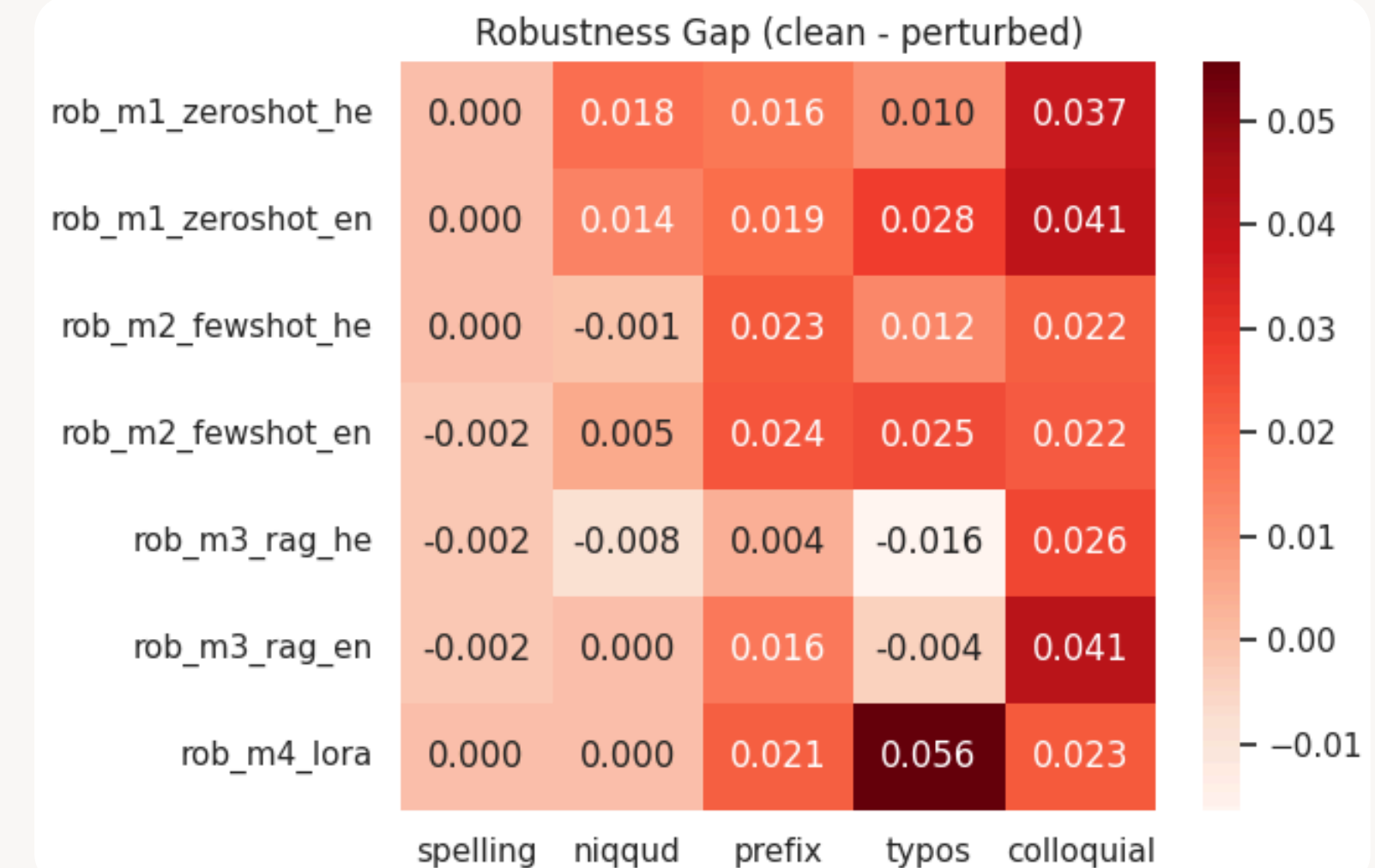
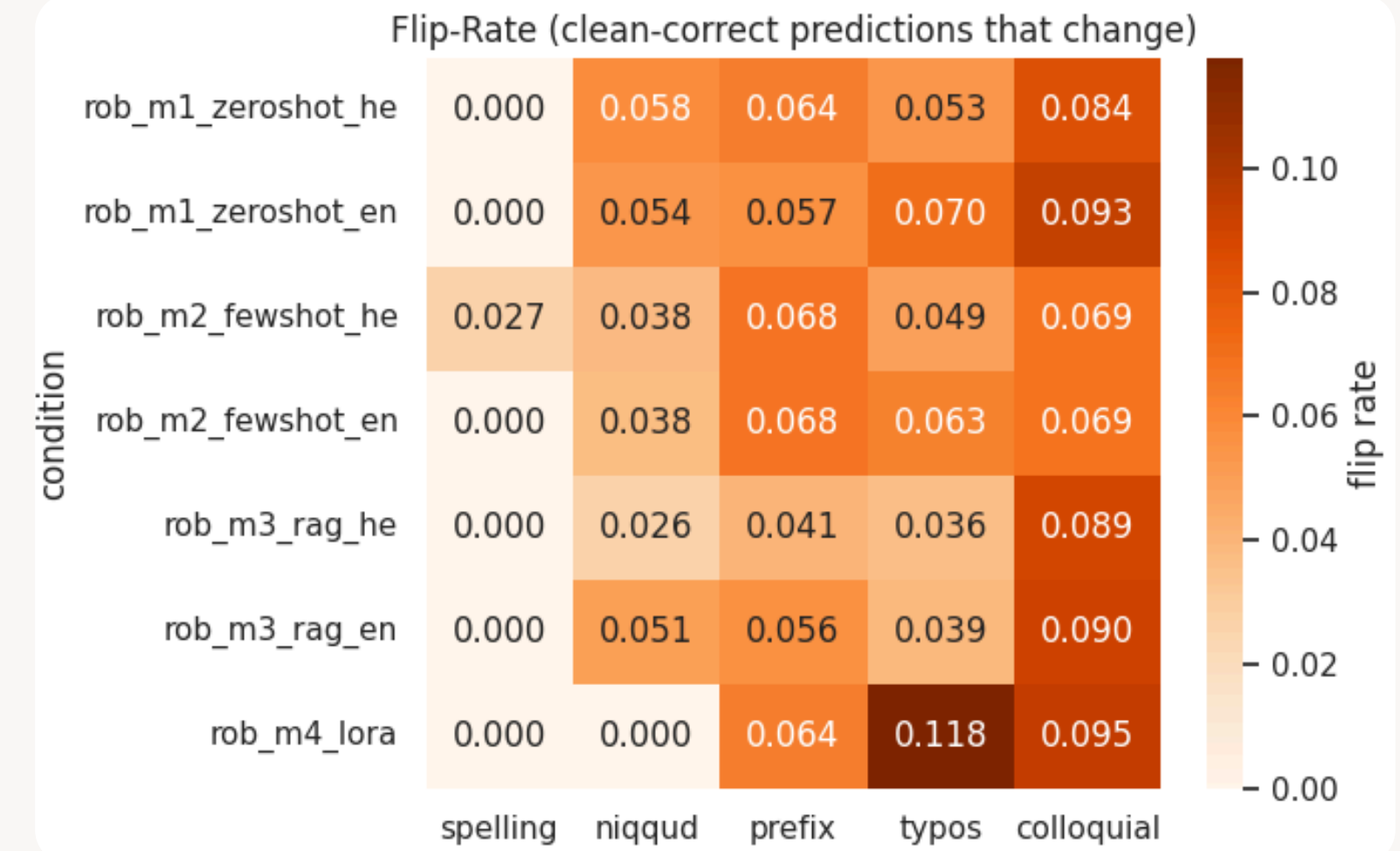
## Best clean Macro-F1:

RAG-EN (0.858) - but clean scores don't predict robustness.



# Finding 1 - retrieval buys robustness

- RAG has the **smallest (often negative) gaps** and **lowest flip-rates** on the rule-based families.
- Retrieving similar Hebrew examples **re-anchors** the model on meaning, not surface form
- Holds even though we retrieve on the **perturbed** text (realistic drift).



# Finding 2 - the LoRA paradox

## Robust

**Invariant** to spelling & niqqud → gap **0.000**, flip **0.000** (Hebrew tokenizer absorbs them).

## Fragile

**Most fragile of all to typos** → gap **0.056**, flip **0.118** (highest in the whole study).

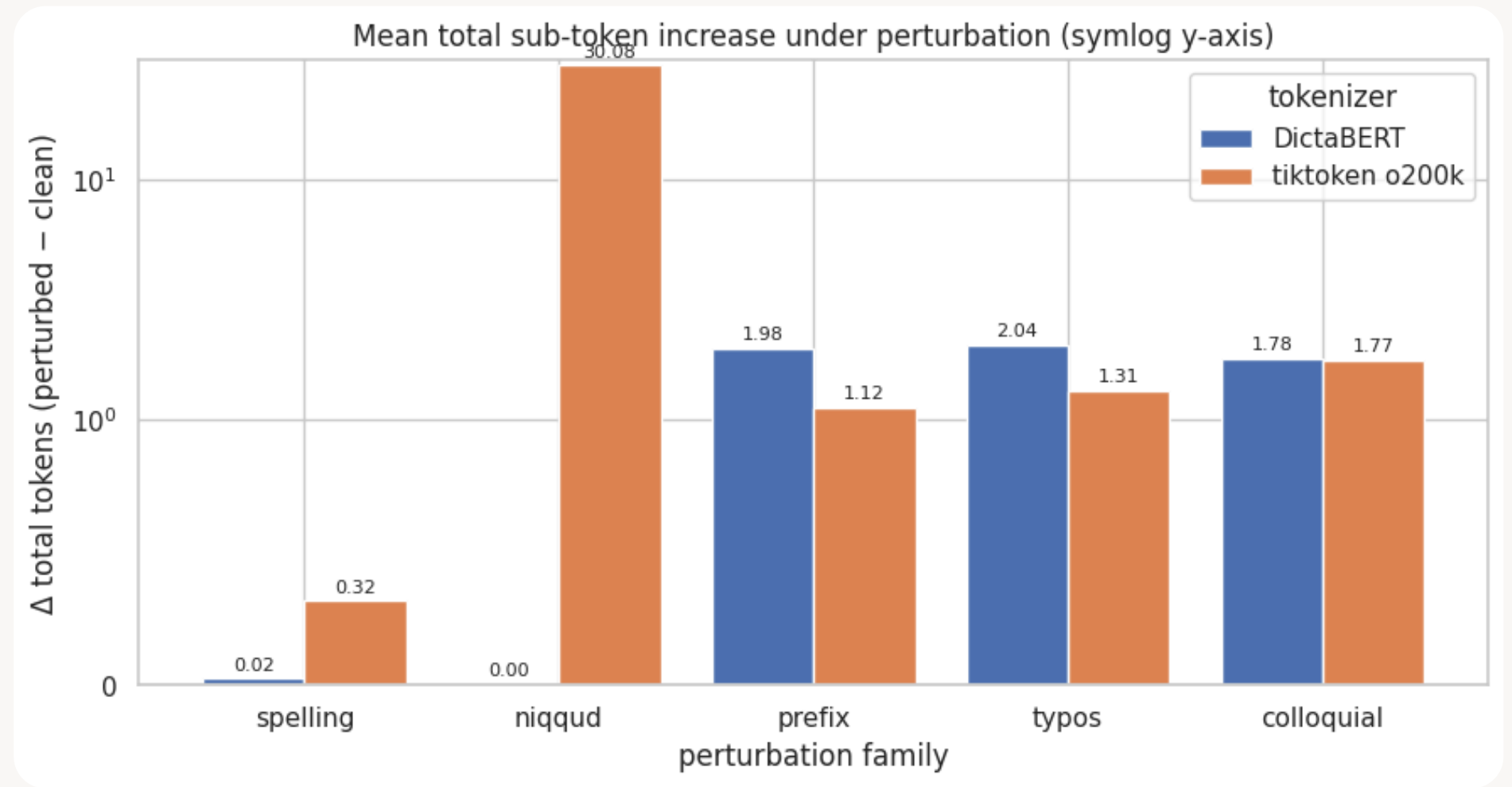
**Why:** final-form normalization (ג→ן, מ→ם) makes words the WordPiece vocab never saw

**A Hebrew tokenizer helps only on what it was trained to normalize**

# Tokenization's Role in Fragility

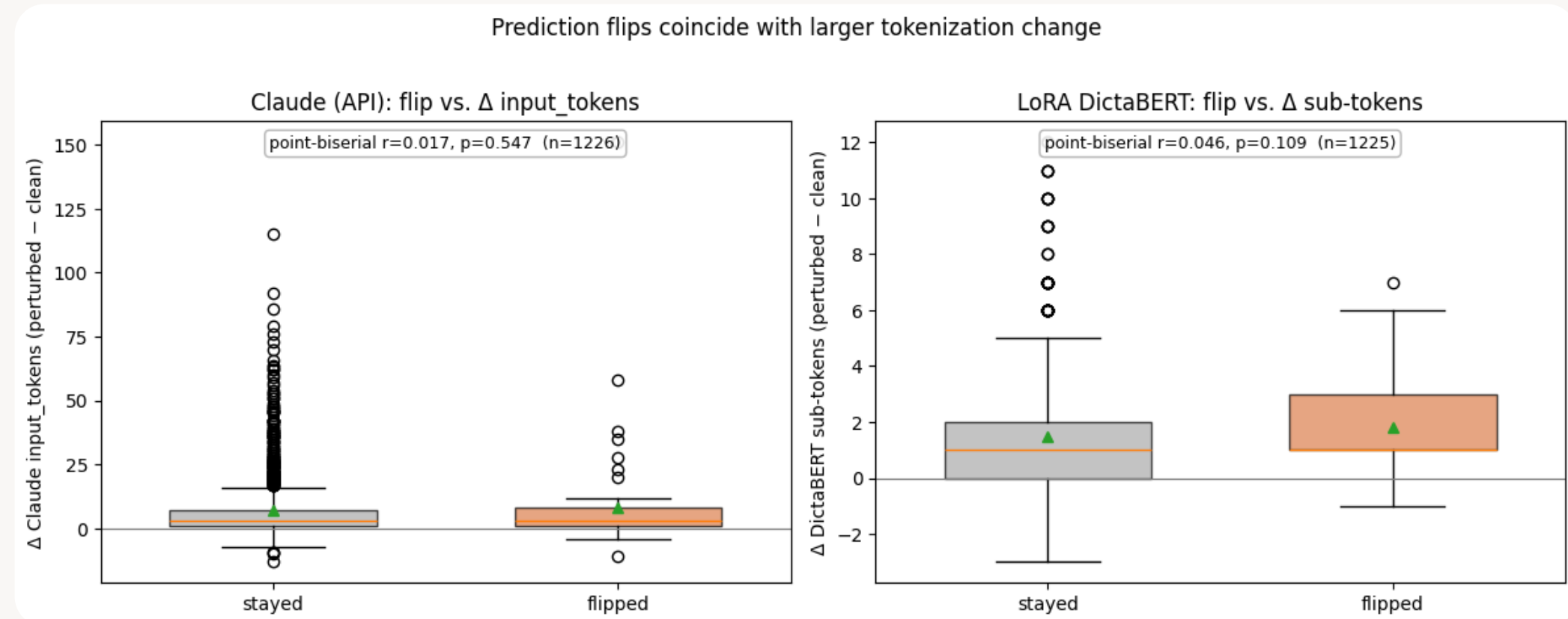
## 01 Niqqud Impact

Niqqud **shatters** the English BPE (+30 tokens) but leaves DictaBERT **unchanged** (0)



## 02 More tokens $\neq$ more fragility

- flips do **not** correlate with token change:  $r \approx 0$ , all  $p > 0.1$  (n.s.)
- Fragility is about **which** sub-tokens appear, not **how many**



# Limitations

## **1. Spelling coverage is small**

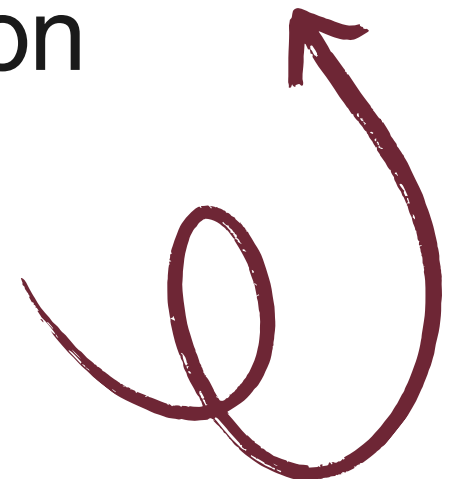
The curated full/defective map only fires on ~8% of examples

## **2. Niqqud needs Dicta Nakdan**

Depends on an external API, so coverage isn't fully under our control.

## **3. Claude's tokenizer is private**

We can't measure its fragmentation directly, so the tokenization analysis leans on tiktoken



# Conclusions

**Q1** What method is most **robust** to meaning-preserving Hebrew variation?

**RAG-augmented prompting is the most robust overall**

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**Q2** Do **Hebrew vs. English** instructions change robustness?

**HE vs. EN instructions: minor next to method choice**

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**Q3** Does **tokenization** explain the fragility?

**Tokenizer fragmentation  $\neq$  fragility.**



# Thank you

## Questions?